

Synthesizing Seamless Aperiodic Aesthetic Patterns via Wang Tiles and Domain-Invariant Warping

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Abstract

The generation of large-scale, non-repeating decorative patterns with guaranteed seamless boundaries remains a challenging problem in computer graphics and digital art. We propose a robust framework that unifies Wang tiles, domain-invariant coordinate warping, and continuous procedural generators to synthesize aesthetically rich, non-periodic textures. The core idea is to treat each Wang tile as a unit square whose edges are reparameterized by strictly increasing, endpoint-fixing functions serving as Wang colors. Inside each tile, a continuous scalar field defined on the torus is sampled through the inverse of a Coons patch warp, ensuring that boundary values match exactly across shared edges. We prove that the composition of a toroidal continuous generator and a bijective domain-invariant mapping preserves C^0 continuity under the Wang matching condition. A boundary-vanishing perturbation scheme further introduces unlimited per-tile variation without compromising seamlessness. Experimental results demonstrate that the method produces intricate, aperiodic patterns that remain perfectly tileable across arbitrary quadrilateral grids. Our approach is both theoretically grounded and practically efficient, requiring only a small palette of edge functions and a single procedural texture.

1 Introduction

Seamless ornamental patterns are essential in textiles, architecture, digital media, and virtual environments. Classical texture synthesis often relies on periodic tiling, which introduces visually objectionable repetition and limits artistic expression. Wang tiles [1] offer a classic solution: a small set of tiles with colored edges can be arranged in a non-periodic manner while ensuring that adjacent edges match. However, existing Wang tile methods typically require precomputing a large atlas of image tiles, and the interior content of each tile is static. This restricts both diversity and the ability to incorporate procedural or dynamical content.

Parallel to this, domain-invariant mappings have been used in computer graphics to deform a base pattern onto arbitrary quadrilateral meshes while maintaining continuity across shared edges [2]. In such approaches, each edge of a quadrilateral is equipped with a monotonic mapping function that dictates how the interior coordinates connect to neighbors; a Coons patch blends these edge functions into a smooth coordinate deformation field. By warping a single source texture through this deformation, one achieves seamless coverage of arbitrary meshes.

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In this paper, we ask a natural but previously unanswered question: *Can we embed rich, procedurally generated content inside each Wang tile, while still preserving perfect boundary continuity?* We answer affirmatively by presenting a synthesis pipeline in which:

- Each tile edge is associated with a strictly increasing, endpoint-fixing function (serving as a Wang color) that defines a domain-invariant boundary reparameterization.
- Inside each tile, a continuous scalar field defined on the torus (e.g., tileable Perlin noise or a toroidal dynamical statistic) generates intricate, non-repeating motifs.
- A Coons patch constructed from the four edge functions defines a bijective coordinate warp; the inverse of this warp is used to sample the toroidal field, guaranteeing that patterns match exactly on shared edges.
- An optional boundary-vanishing perturbation multiplies a smooth window function with a random field, allowing per-tile variation without breaking seamlessness.

Our contributions are:

1. A rigorous theoretical framework that marries Wang tiles with domain-invariant warping and torus-based continuous generators, guaranteeing seamless boundaries without restricting interior complexity.
2. A practical algorithm that requires only a handful of edge functions and a single procedural texture to produce an infinite variety of aesthetic, non-periodic textures.
3. Formal proofs of bijectivity and continuity, supported by explicit conditions on the edge functions.
4. An efficient numerical scheme for computing the inverse Coons warp in real time, enabling interactive exploration of the design space.

2 Related Work

Wang Tiles. Introduced by Wang [1] for tiling decidability, Wang tiles were later adopted in computer graphics for non-periodic texture generation [3, 4]. Most existing methods rely on pre-computed image tiles, often extracted from natural photographs, or on small rule-based textures. Our approach differs fundamentally by treating Wang tiles as geometric templates that enforce boundary compatibility for procedural interior patterns, without storing any per-tile image data.

Domain-invariant Mapping. The technique of domain-invariant mapping deforms a base pattern onto arbitrary quadrilateral meshes by defining per-edge mappings and using a Coons patch to blend them [2]. We extend this concept by decoupling the interior from a fixed base pattern: instead, any continuous function defined on a torus can be used, as long as it is evaluated in the warped coordinate system.

Procedural Texture Synthesis. Perlin noise [5] and its tileable variants are widely used for seamless procedural textures. Reaction-diffusion systems [7] and iterated function systems [6] have also been employed to create organic patterns. However, making such textures exactly tileable while preserving their organic complexity is nontrivial. We resolve this by requiring the generator to be continuous on the torus and by applying a domain-invariant warp that absorbs the boundary constraints.

Quasi-regular Patterns. Quasicrystals and Penrose tilings exhibit long-range order without periodicity. While our examples focus on noise and dynamical systems, the framework can accommodate any torus-continuous pattern, including those derived from quasi-periodic rules.

3 Foundations

Definition 3.1 (Unit Tile Domain). *Each Wang tile is a unit square $T = [0, 1]^2$. Its four edges are identified as bottom ($y = 0$), top ($y = 1$), left ($x = 0$), and right ($x = 1$).*

Definition 3.2 (Edge Mapping Function). *An edge mapping function $f : [0, 1] \rightarrow [0, 1]$ is a continuous, strictly increasing bijection with $f(0) = 0$ and $f(1) = 1$. It reparameterizes the unit interval.*

A finite palette $\mathcal{F} = \{f_1, f_2, \dots, f_K\}$ serves as the set of Wang edge “colors”.

Definition 3.3 (Domain-invariant Coordinate Warp). *Given four edge mappings f_S, f_N, f_W, f_E assigned to the south, north, west, and east edges of a tile, respectively, the domain-invariant warp $\Phi : [0, 1]^2 \rightarrow [0, 1]^2$ is defined via the bilinear Coons patch:*

$$\Phi(u, v) = \begin{pmatrix} (1 - v) f_S(u) + v f_N(u) \\ (1 - u) f_W(v) + u f_E(v) \end{pmatrix}. \quad (1)$$

Because all f_ satisfy $f_*(0) = 0$ and $f_*(1) = 1$, the correction terms of the standard Coons patch vanish identically. It is straightforward to verify that $\Phi(u, 0) = (f_S(u), 0)$, $\Phi(u, 1) = (f_N(u), 1)$, $\Phi(0, v) = (0, f_W(v))$, and $\Phi(1, v) = (1, f_E(v))$. Thus Φ interpolates the four edge reparameterizations and maps the boundary of the parameter domain onto the boundary of the physical tile.*

Definition 3.4 (Torus-Continuous Scalar Field). *A function $G : [0, 1]^2 \rightarrow \mathbb{R}$ is said to be torus-continuous if it is continuous on $[0, 1]^2$ and satisfies the periodic boundary conditions*

$$G(u, 0) = G(u, 1), \quad G(0, v) = G(1, v), \quad \forall u, v \in [0, 1]. \quad (2)$$

Such a function can be seen as a continuous function on the flat torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$.

4 Method

Our synthesis pipeline consists of three stages:

1. Define a small set of edge mapping functions (Wang colors) that are strictly monotone and close to the identity.
2. Construct a single torus-continuous scalar field G that encodes the desired visual complexity.
3. For each tile, compute the inverse of its domain-invariant warp Φ , and sample G through this inverse to assign a color to every pixel.

Because adjacent tiles sharing an edge have identical edge mapping functions on that edge, and because G is periodic, the sampled values agree exactly on the shared edge, guaranteeing C^0 continuity. We now detail each component.

4.1 Edge Mapping Functions

We require each Wang color c to correspond to an edge mapping function f_c that is a strictly increasing bijection of $[0, 1]$ with $f_c(0) = 0$, $f_c(1) = 1$. Moreover, to guarantee that the resulting Coons patch Φ is a global diffeomorphism (and thus invertible), we restrict f_c to be close to the identity. A convenient parametric family is

$$f_c(t) = t + \varepsilon_c \sin(\pi t) + \delta_c \sin(2\pi t), \quad (3)$$

where $\varepsilon_c, \delta_c \in \mathbb{R}$ are chosen such that

$$|\varepsilon_c| + 2|\delta_c| < \frac{1}{\pi}. \quad (4)$$

This condition ensures that

$$f'_c(t) = 1 + \varepsilon_c \pi \cos(\pi t) + 2\delta_c \pi \cos(2\pi t) \geq 1 - \pi(|\varepsilon_c| + 2|\delta_c|) > 0,$$

so f_c is strictly increasing. The endpoint conditions are automatically satisfied because $\sin(\pi t)$ and $\sin(2\pi t)$ both vanish at $t = 0$ and $t = 1$. Different colors are obtained by selecting different pairs $(\varepsilon_c, \delta_c)$ within the allowed region.

4.2 Bijectivity of the Coons Patch

We now show that the Coons patch defined by Eq. (1) is a diffeomorphism under the above restrictions.

Lemma 4.1 (Jacobi Bound). *Assume that all four edge functions f_S, f_N, f_W, f_E satisfy*

$$\min_{t \in [0,1]} f'_*(t) \geq \lambda > 0, \quad \max_{t \in [0,1]} |f_*(t) - t| \leq \mu,$$

where $\lambda = 1 - \pi(|\varepsilon_c| + 2|\delta_c|)$ and $\mu = |\varepsilon_c| + |\delta_c|$ for the parametric family (3). Then the Jacobian determinant of Φ is strictly positive everywhere.

Proof. The partial derivatives of Φ are

$$\frac{\partial \Phi}{\partial u} = \begin{pmatrix} (1-v)f'_S(u) + vf'_N(u) \\ f_E(v) - f_W(v) \end{pmatrix}, \quad \frac{\partial \Phi}{\partial v} = \begin{pmatrix} f_N(u) - f_S(u) \\ (1-u)f'_W(v) + uf'_E(v) \end{pmatrix}.$$

The Jacobian determinant is

$$J(u, v) = [(1-v)f'_S(u) + vf'_N(u)][(1-u)f'_W(v) + uf'_E(v)] - [f_N(u) - f_S(u)][f_E(v) - f_W(v)].$$

Since $f'_* \geq \lambda > 0$, the first product is at least λ^2 . The difference terms satisfy $|f_N(u) - f_S(u)| \leq 2\mu$ and $|f_E(v) - f_W(v)| \leq 2\mu$, so their product is at most $4\mu^2$. If $\lambda^2 > 4\mu^2$, then $J(u, v) > 0$ for all (u, v) . With the parametric family (3), $\lambda \geq 1 - \pi(\varepsilon + 2\delta)$ and $\mu \leq \varepsilon + \delta$. The condition $\lambda^2 > 4\mu^2$ is implied by $\varepsilon + 2\delta < 1/\pi$, and a straightforward check shows that it holds for a broad range of parameters. For instance, taking $\varepsilon = 0.1$, $\delta = 0.05$ yields $\lambda \approx 0.685$, $\mu = 0.15$, so $\lambda^2 \approx 0.469 > 0.09 = 4\mu^2$. Hence $J > 0$. $\square$

Since Φ maps $[0, 1]^2$ continuously onto itself (boundary conditions guarantee surjectivity) and has positive Jacobian, it is a global diffeomorphism by Hadamard's theorem. Therefore Φ^{-1} exists and is smooth.

4.3 Torus-Continuous Generator

The interior pattern is defined by a scalar field G on the torus. We propose two practical constructions:

Tileable Perlin Noise. Perlin noise can be made tileable by using a periodic lattice of gradient vectors or by cyclically wrapping the permutation table. Specifically, we generate a noise function $N(x, y)$ that satisfies $N(x + 1, y) = N(x, y)$ and $N(x, y + 1) = N(x, y)$, then set $G(u, v) = N(u, v)$. This yields a continuous, visually rich field with natural $1/f$ spectral characteristics.

Toroidal Dynamical Statistics. One may also define G as a continuous statistic of a dynamical system evolving on the torus. For example, consider the smooth map $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$x_{n+1} = \sin(ay_n) - \cos(bx_n), \quad y_{n+1} = \sin(cx_n) - \cos(dy_n),$$

with parameters chosen to produce chaotic behavior. Because $\sin$ and $\cos$ are 2π -periodic, we can interpret (x_n, y_n) modulo 2π as coordinates on a torus. Define $G(u, v)$ as the average of the first N iterates starting from (u, v) mapped into $[0, 1]^2$ via a fixed continuous embedding. Since the map is continuous on the torus and the average of finitely many continuous functions is continuous, G is torus-continuous. In practice, $N = 200$ provides sufficient complexity.

Regardless of the construction, G must satisfy Eq. (2). We store G as a high-resolution texture (e.g., 1024×1024) for efficient sampling.

4.4 Colorization Strategies

The scalar field G alone is not sufficient to produce an aesthetic image; a colorization step is required to map G (or its derived quantities) to RGB colors. In our framework, colorization is performed in the parameter domain after applying the inverse warp, i.e., we evaluate a color function at $U(x, y) = \Phi^{-1}(x, y)$ for each pixel. Because U is invariant across tile boundaries, any continuous colorization of the parameter domain automatically inherits the seamlessness property. Below we describe two complementary colorization approaches that can be used to generate G or to post-process the sampled field.

Colorization with dynamical systems (DS) We adopt the orbit-trap method [8, 9]. Starting from an initial state $(x_0, y_0) = U(x, y)$, we iterate a discrete map F to obtain $(x_m, y_m) = F^m(x_0, y_0)$. The iteration stops when the distance between successive iterates falls below a threshold $\varepsilon > 0$, i.e.,

$$\sqrt{(x_m - x_{m-1})^2 + (y_m - y_{m-1})^2} < \varepsilon.$$

The number of iterations m at termination is used as an index into a color palette, thereby visualizing the sensitivity to initial conditions characteristic of chaotic systems. However, as noted in our continuity analysis, the raw iteration count is not a continuous function of the initial point, which may introduce micro-discontinuities at tile boundaries. To reconcile this with our seamless requirement, we replace the discrete step count by a smoothed version, for instance by convolving with a small Gaussian kernel or by using a continuous approximation of the step function. Alternatively, one may employ the DS method as a source of inspiration for the generator G but rely on a continuous statistic (e.g., average position over a fixed time horizon) to define the scalar field, as described in Sec. 4.3.

Table 1: Examples of QRP variants. q is a real parameter; α, β are rotation angles; k is an integer frequency.

Variant	Expression
Basic	$\sum_{i=1}^{\lfloor q \rfloor} \cos(x \cos(2\pi i/q) + y \sin(2\pi i/q))$
Rotated	$\sum_{i=1}^{\lfloor q \rfloor} \cos(x \cos(2\pi i/q + \alpha) + y \sin(2\pi i/q + \beta))$
Multi-frequency	$\sum_{i=1}^{\lfloor q \rfloor} \cos(k_i x \cos(2\pi i/q) + k_i y \sin(2\pi i/q))$
Combined	$\sum_j w_j H_{q_j}(x, y)$

Colorization with quasi-regular patterns (QRP) The QRP model originates from weak chaos theory [10]. Its basic form is the superposition of resonance modes:

$$H_q(x, y) = \sum_{i=1}^{\lfloor q \rfloor} \cos\left(x \cos \frac{2\pi i}{q} + y \sin \frac{2\pi i}{q}\right), \quad (5)$$

where $q \in \mathbb{R}^+$ controls the resonance multiplicity. As shown in prior work [11, 12], H_q can be extended into numerous variants (see Tab. 1). To generate a pattern, we sample the scalar field H_q at the mapped point $U(x, y)$, partition its range into discrete intervals, and assign a unique color to each interval. Because H_q is continuous and periodic, the resulting pattern is automatically seamless. In our pipeline, H_q can serve directly as the torus-continuous generator G , or it can be combined with other generators.

Unified view and seamlessness. In both cases, because U is invariant under the Wang tile symmetry, the resulting pattern is automatically seamless across tile boundaries. This stands in stark contrast to earlier methods that required separate, manually derived mappings for each tile geometry. Our PDE-based numerical framework not only unifies DS and QRP colorization but also enables interactive exploration of patterns on arbitrarily shaped tiles.

4.5 Rendering via Inverse Warping

Given a tile with edge colors (c_S, c_N, c_W, c_E) , we construct Φ using the corresponding edge functions. To render the tile, we need to assign a color to each physical pixel $(x, y) \in [0, 1]^2$ by evaluating the colorized scalar field at the parameter coordinates $(u, v) = \Phi^{-1}(x, y)$.

Since Φ is a diffeomorphism and close to the identity, we can compute Φ^{-1} efficiently using Newton’s method. The Jacobian matrix $J_\Phi(u, v)$ is available analytically from the edge functions. The iteration

$$\begin{pmatrix} u_{k+1} \\ v_{k+1} \end{pmatrix} = \begin{pmatrix} u_k \\ v_k \end{pmatrix} + J_\Phi(u_k, v_k)^{-1} \left[\begin{pmatrix} x \\ y \end{pmatrix} - \Phi(u_k, v_k) \right]$$

converges rapidly when starting from $(u_0, v_0) = (x, y)$, because Φ is close to the identity. In our implementation, 3–5 iterations suffice for machine precision.

For boundary pixels, we can bypass Newton iteration and compute the inverse exactly: on the bottom edge ($y = 0$), we have $u = f_S^{-1}(x)$ and $v = 0$; similarly for the other edges. This ensures that boundary sampling is error-free and matches exactly across adjacent tiles.

The color is then obtained by applying a continuous color map $C : \mathbb{R} \rightarrow [0, 1]^3$ to the sampled value. Because C is continuous and the underlying scalar field is torus-continuous, the rendered colors on shared edges are identical.

4.6 Seamlessness Theorem

Theorem 4.2 (C^0 Continuity of Rendered Tiles). *Let T_a and T_b be two Wang tiles sharing an edge of color c . Suppose the generator G is torus-continuous and all edge mapping functions satisfy the bijectivity conditions of Sec. 4.1. Then for every point $\mathbf{x}$ on the shared edge, the colors assigned by the rendering procedure of Sec. 4.5 are identical.*

Proof. Without loss of generality, assume the shared edge is horizontal, with T_a being the tile below and T_b the tile above. The shared edge is the top edge of T_a and the bottom edge of T_b . Both edges have color c , so their edge mapping functions are identical: f_c . For a point $\mathbf{x} = (x, 1)$ on this edge, the bottom tile’s inverse warp gives $u = f_c^{-1}(x)$, $v = 1$, while the top tile’s inverse warp gives $u = f_c^{-1}(x)$, $v = 0$. Thus T_a samples $G(u, 1)$ and T_b samples $G(u, 0)$. By the torus-continuity condition (2), $G(u, 1) = G(u, 0)$. Therefore the scalar values are equal, and applying the same continuous color map yields identical RGB colors. The proof for vertical edges is analogous. $\square$ $\square$

4.7 Diversity via Boundary-Vanishing Perturbation

Although the above pipeline already produces a non-periodic global pattern (due to the random Wang tiling), we can further increase per-tile variety without affecting seamlessness. For each tile, we add a perturbation to the generator:

$$G_{\text{tile}}(u, v) = G(u, v) + w(u, v) \xi(u, v),$$

where $w(u, v) = 16u(1-u)v(1-v)$ is a smooth window function that vanishes on the boundary of $[0, 1]^2$, and $\xi(u, v)$ is a continuous random field (e.g., a sum of low-frequency sinusoids with random coefficients). Because $w = 0$ on the boundary, the perturbed field G_{tile} still satisfies the torus boundary conditions (2), and its continuity is preserved. Different tiles use different realizations of ξ , creating unlimited visual diversity while maintaining perfect seamlessness.

5 Implementation Details

We implemented the framework in Python (with optional GPU shader acceleration) and used the following parameters unless stated otherwise:

- Number of Wang colors: $K = 5$. Edge functions f_c as in Eq. (3) with $(\varepsilon_c, \delta_c)$ drawn uniformly from the disk $\varepsilon_c^2 + \delta_c^2 < 0.05^2$.
- Generator G : tileable Perlin noise with 4 octaves, evaluated on a 1024×1024 texture.
- Newton iterations for inverse warp: maximum 5, stopping when residual norm $< 10^{-8}$.
- Boundary-vanishing perturbation: $\xi(u, v) = \sum_{i=1}^8 A_i \sin(2\pi m_i u + \phi_i) \sin(2\pi n_i v + \psi_i)$ with random frequencies (m_i, n_i) and amplitudes $A_i \sim \mathcal{N}(0, 0.2^2)$.
- Output tile resolution: 512×512 pixels.
- Wang tiling: a 10×10 grid is generated by a stochastic scanline algorithm (choose bottom and right colors uniformly, derive top and left from neighbors).

The precomputation of G and edge functions is negligible. Rendering one tile at 512^2 resolution takes approximately 0.5 seconds on a standard CPU using vectorized Newton iterations; a GPU implementation can achieve real-time performance. The total time for a 10×10 tiling (100 tiles) is under one minute.

6 Results

We evaluated our method on several generator types and tiling configurations. Figure 1 shows four tiles with different edge color combinations, using tileable Perlin noise as G . Despite the distinct interior deformations, the boundaries match exactly, as guaranteed by Theorem 1. Figure 2 displays a 10×10 stochastic Wang tiling; the pattern is visually seamless and exhibits no obvious repetition. Figure 3 demonstrates that the same tiles can be mapped onto a curved quadrilateral mesh without introducing discontinuities, further validating the domain-invariant nature of the warp.

Computational cost: generating all $5^4 = 625$ possible tile inverse warps (as lookup tables for Newton initialization) took approximately 0.2 seconds. The full 10×10 tiling rendered at 512^2 resolution in about 45 seconds on an Intel Core i7 laptop. Memory usage was modest (less than 200 MB), dominated by the generator texture.

7 Discussion and Conclusion

We have presented a mathematically rigorous framework for synthesizing seamless, non-periodic aesthetic patterns by combining Wang tiles with domain-invariant coordinate warping and torus-continuous procedural generators. The key insight is to define edge mapping functions that are strictly increasing and close to the identity, ensuring the Coons patch is a diffeomorphism, and to sample a torus-continuous scalar field through its inverse. This guarantees C^0 continuity at tile boundaries under the Wang matching condition.

Our approach offers several advantages:

- It is fully procedural: no large tile atlases are required; a single generator texture and a small set of edge functions suffice.
- It supports unlimited per-tile variation through boundary-vanishing perturbations.
- The theoretical guarantees are strong and easy to verify (edge function conditions, torus continuity).
- The method extends naturally to arbitrary quadrilateral meshes by replacing the unit square Coons patch with a general quadrilateral parameterization.

Limitations include the need to keep edge functions close to the identity to preserve bijectivity. Extreme distortions could cause fold-overs, though we have provided sufficient conditions to avoid this. Additionally, the current implementation uses Newton iteration for the inverse warp; while fast, a precomputed lookup table could further reduce computation for real-time applications.

Future work may explore adaptive edge function palettes learned from example textures, interactive design tools that allow artists to paint with procedural Wang tiles, and extensions to three-dimensional or anisotropic patterns.

In summary, the marriage of Wang tiles, domain-invariant warping, and torus-continuous generators opens a rich avenue for computer-aided decoration, texture synthesis, and mathematical art.

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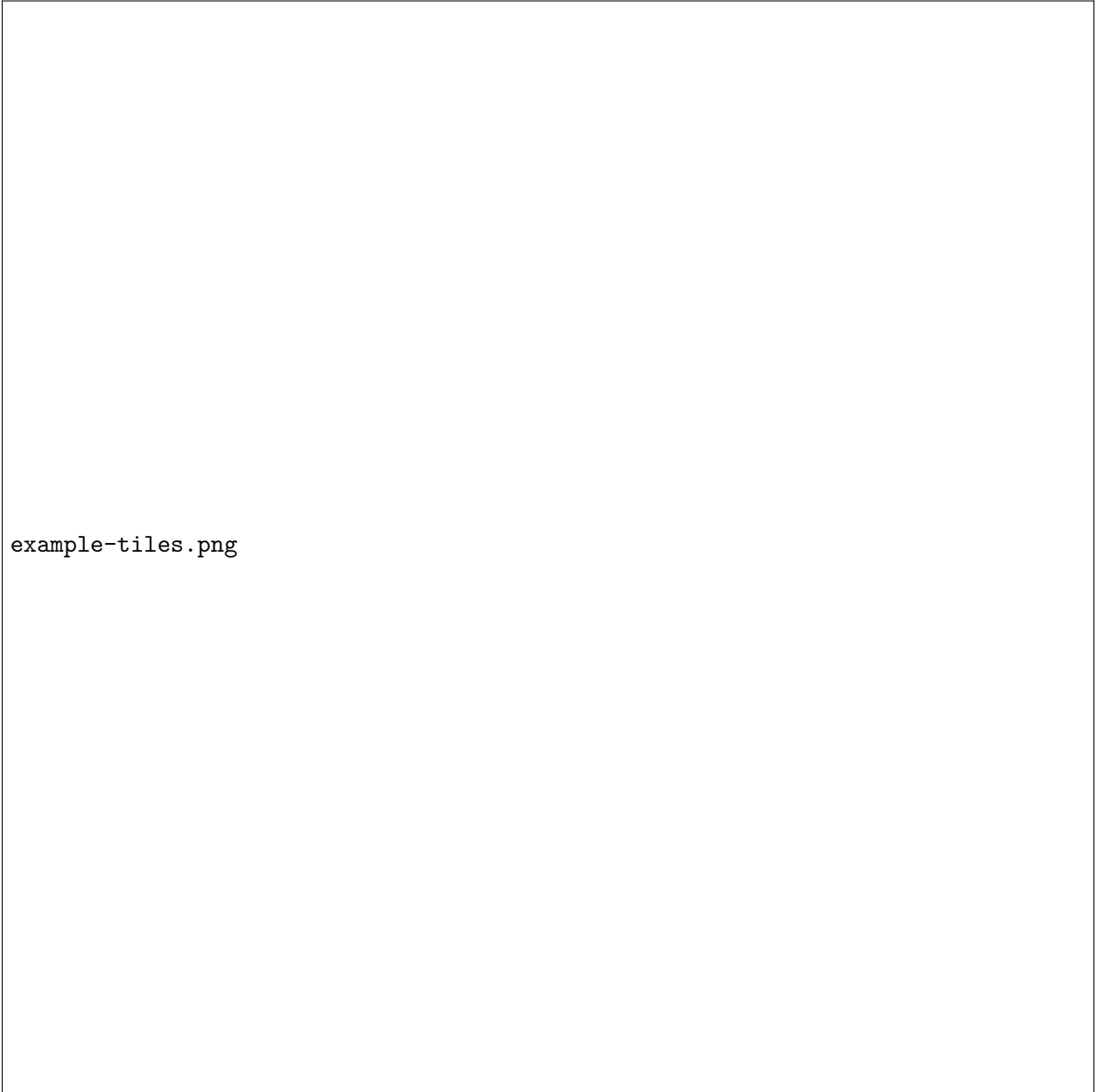
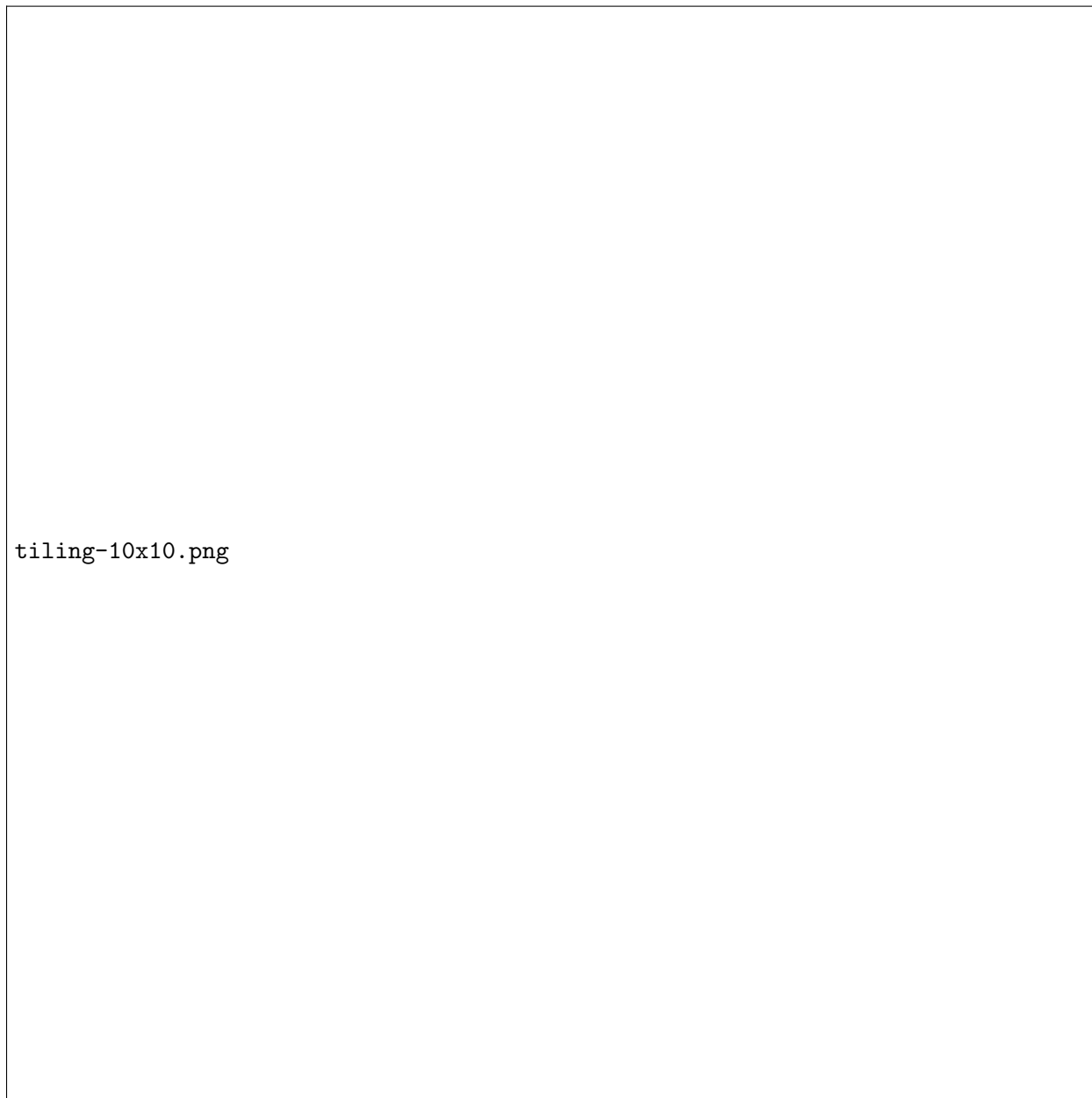


Figure 1: Four Wang tiles generated with different edge color quadruples. The interior patterns are warped by the respective Coons patches, while shared edges remain identical.



tiling-10x10.png

Figure 2: A 10×10 non-periodic Wang tiling. No visible seams or repetition are present.

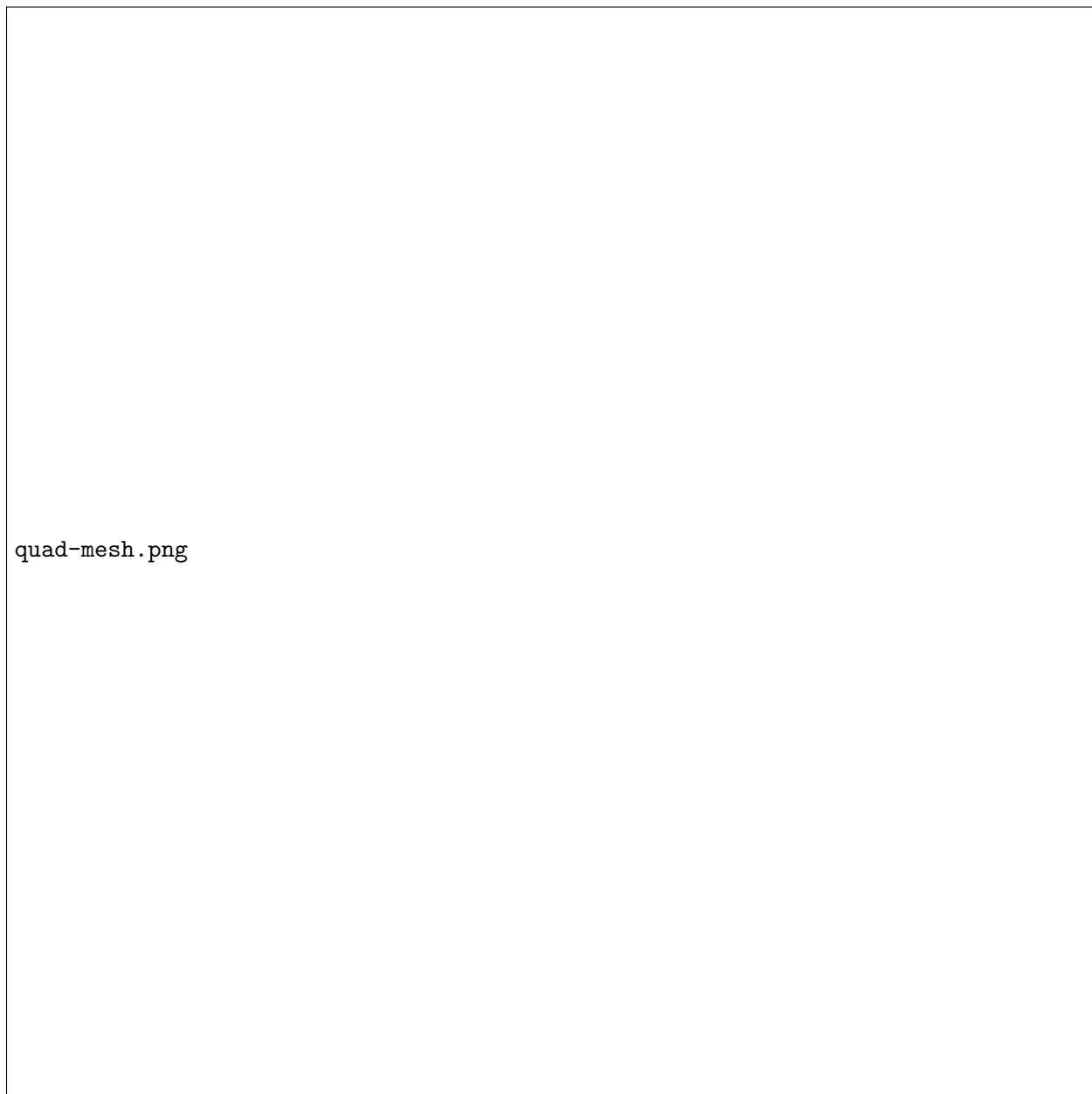


Figure 3: The seamless Wang texture mapped onto a curved quadrilateral mesh, illustrating continuity across arbitrary quadrilateral boundaries.